

GOKUL KRISHNAN G

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linkedin.com/in/gokul-krishnan-g-2026-ai | github.com/GOKULKRISHNAN2044 | Chennai, India (Open to Relocate / Remote)

PROFESSIONAL SUMMARY

AI-Native Software Engineer with **1 year cumulative engineering experience** (including **8+ months production AI** at AVASOFT) and a **B.Tech in AI & Data Science (8.33 CGPA)**. Proven track record in architecting **Autonomous Multi-Agent Systems (LangGraph, FastMCP)**, **Vision LLM Pipelines (AWS Bedrock)**, and **Concurrent Microservices (Go 1.22+, FastAPI, pgvector)** with strict 4-tier Clean Architecture.

TECHNICAL SKILLS

Languages: Python, Go (Golang 1.22+), Java, JavaScript (ES6+), SQL, HTML5, CSS3

AI & Agentic Systems: LangGraph, LangChain, FastMCP (Model Context Protocol), MultiServerMCPClient, RAG, HITL

LLMs & Cloud AI: Google Gemini AI, AWS Bedrock (Claude Haiku 4.5 Vision, Titan Embeddings), OpenAI APIs, Ollama, SBERT

Computer Vision & ML: OpenCV, TensorFlow, Keras CNN, Scikit-Learn, OCR Text Extraction, Image Preprocessing

Backend & Architecture: Clean/Layered Architecture, Gorilla Mux, FastAPI, GORM ORM, SQLAlchemy 2.0, REST APIs

Databases & Vector Search: PostgreSQL, Supabase, pgvector (Cosine Similarity), MySQL, SQLite, Asyncpg, AsyncPostgresSaver

Cloud & DevOps: AWS (ECS, EventBridge, SES, S3, RDS, CloudWatch, VPC), Docker, Azure DevOps CI/CD, Git, Linux

PROFESSIONAL EXPERIENCE

AVASOFT

Dec 2025 – Aug 2026

AI Analyst / Software Engineer (Full-Time — Joined Mid Final Year)

Chennai, India

Project 1: AI-Powered Ticketing & Multi-Modal Automation Platform

Live Sports & Music Enterprise Client

- Engineered event-driven microservices (**FastAPI & Go, 50+ REST APIs**) processing **30,000+ monthly emails**; slashed incident response times from **4+ hours to <45 seconds** via AWS EventBridge, SES, and ECS.
- Built a **multi-modal AI vision pipeline** using **Claude Haiku 4.5 (AWS Bedrock)** and **OpenCV** OCR to parse emails and attachments with **>97% entity extraction accuracy**.
- Configured **Azure DevOps CI/CD pipelines** with automated builds, tests, and container deployments on PR events.

Project 2: T3 – AI Assessment & Developer Evaluation Platform

AVASOFT Internal Core Product

- Voice AI Interviews:** Built an agentic pipeline with real-time speech transcription and LLM-based rubric candidate scoring.
- MCQ & Live Coding Engine:** Automated question generation; built AST-driven syntax and logic validation via FastAPI.

ENTERPRISE TECHNICAL PROJECTS — Core Systems & Agent Architectures

Enterprise AI Request Triage & Incident Routing System (Client Deliverable) | Google Gemini AI · FastAPI · React · Supabase · PBKDF2 GitHub

- Engineered and delivered an intelligent triage platform for unstructured enterprise requests (support, billing, outages, leads); leveraged **Google Gemini AI** for real-time summarization, 6-category classification, explainable urgency scoring, and automated routing. Built with a **React** dashboard, **salted PBKDF2 authentication**, and **Supabase PostgreSQL** persistence; tested and delivered to client.

Enterprise Room Booking Management Microservice (Go Flagship) | Go 1.22+ · Clean Architecture · GORM · Uber Zap · Docker · Swagger GitHub

- Engineered a high-concurrency booking microservice in **Go 1.22+** implementing strict **4-Layer Clean Architecture** (Handlers, Services, Repositories, Entities), GORM transactional persistence, Uber Zap logging, Docker, and Swagger docs.

FastAPI Clean Architecture REST API | Python 3.11+ · FastAPI · SQLAlchemy 2.0 · PostgreSQL · Pytest · UV GitHub

- Architected a modular REST API following strict **Clean Layered Architecture** principles (Routers → Services → Repositories → Models) with Pydantic v2 schemas, async PostgreSQL, and automated Pytest/UV test suites.

Autonomous Multi-Agent Code Review System & FastMCP Framework | LangGraph · AWS Bedrock · FastMCP · AST Analysis · Python LangGraph

- Orchestrated stateful collaborative agents in **LangGraph** with cyclic self-healing AST linting, AsyncPostgresSaver persistent memory, and **FastMCP Human-in-the-Loop** decision checkpoints (place_order) via MultiServerMCPClient.

Enterprise RAG Knowledge Base & Semantic Recommendation System | FastAPI · AWS Bedrock · pgvector · Cosine Search · Python GitHub

- Architected a 2-stage asynchronous RAG pipeline: automated document ingestion, semantic chunking, high-dimensional vector embeddings, and Top-K cosine distance retrieval over **PostgreSQL pgvector** with hallucination-free LLM synthesis.

INTERNSHIPS

Edu-Varsity <i>Cloud Computing Intern</i> Role Definition: Architected and deployed scalable, isolated multi-tier cloud infrastructure on AWS for document processing with automated health telemetry. Project: Cloud-Native Resume Enhancer Web Platform - Deployed multi-tier containers on AWS ECS inside a custom VPC with isolated public/private subnets and RDS PostgreSQL. - Engineered S3 pre-signed upload pipelines, CloudWatch health alarms, and multi-stage Docker container image optimization.	Jun 2024 – Jul 2024 Chennai, India <i>Tech:</i> AWS (VPC, ECS, RDS, CloudWatch, S3, IAM), Docker
FLASH INFOLABS <i>Full Stack Developer Intern</i> Role Definition: Engineered full-stack REST API microservices, database schemas, and interactive user directories for corporate workforce operations. Project: Enterprise Employee Management & Directory Portal <i>Tech:</i> Python FastAPI, PostgreSQL, JavaScript, HTML5, CSS3, REST APIs - Built modular backend REST APIs in Python FastAPI with Pydantic v2 schema validations and relational database queries. - Implemented RBAC access security, standardized JSON responses, and responsive single-page search interfaces via modern JavaScript.	Oct 2023 – Nov 2023 Chennai, India

ACADEMIC & RESEARCH CAPSTONES — University Engineering Foundations

AI-Powered Psychometric Assistant for Stress <i>Python · SBERT · DistilBERT · Gemini API · Rasa</i> - Developed an adaptive stress evaluation assistant using SBERT sentence embeddings, DistilBERT classification, and Gemini API tone analysis integrated with an interactive Rasa conversational chatbot. - Engineered a real-time stress assessment dashboard featuring speedometer visualizations and question sequencing.	 GitHub
Automated Guided Vehicle (AGV) — Intelligent Navigation <i>Python · OpenCV · Path Planning · NumPy</i> - Engineered a computer vision trajectory planning and autonomous navigation system for automated guided vehicles using OpenCV image processing and NumPy path planning algorithms. - Implemented real-time obstacle avoidance and edge tracking; open-source codebase forked by 2 external community contributors.	 GitHub
Presence Pro — Face Recognition & Biometric Attendance <i>Python · OpenCV · Haar Cascades · SQLite</i> - Implemented real-time biometric face verification using OpenCV Haar Cascades and 128-d face encodings linked to SQLite. - Automated absentee notifications via SMTP email protocols with timestamped attendance audit logs and summary reports.	 GitHub

ACHIEVEMENTS & RESEARCH PUBLICATIONS

 1st Prize — Ideathon RESONANCE '25, Rajalakshmi Engineering College (IEEE Computer Society Chapter)	Feb 2025
 2 International Research Publications IJFMR (IF: 9.24, 2025) & TIJER (IF: 8.57, Jul 2024)	2024 – 2025
 IEEEExtreme 18.0 Global programming competition, ranked among 19,000+ participants	Oct 2024

EDUCATION

Panimalar Engineering College <i>B.Tech in Artificial Intelligence & Data Science (CGPA: 8.33 / 10)</i>	2022 – 2026 Chennai, India
Jayam Metric Higher Secondary School <i>Higher Secondary Certificate — Computer Science (88.34%)</i>	May 2022 Dharmapuri, India

KEY CERTIFICATIONS

 Oracle Cloud Infrastructure 2024 Generative AI Certified Professional Oracle University	Jul 2024 – Jul 2026
 NPTEL — Introduction to Internet of Things IIT Kharagpur / SWAYAM (Elite Certificate)	2024
 Machine Learning Training AcmeGrade × IIT Bombay Mood Indigo	2023